

Ans 1. Gradient of a field is referred to

$$\nabla f = \frac{\partial f}{\partial x_1} \hat{n}_1 + \frac{\partial f}{\partial x_2} \hat{n}_2 + \frac{\partial f}{\partial x_3} \hat{n}_3$$

$$\Rightarrow \nabla (x+2y+3z) = (1, 2, 3)$$

$$\Rightarrow \nabla (x+2y+3z)_{\text{At } (1,2,3)} = \nabla (x+2y+3z)_{\text{At } (4,-2,9)} \\ = (1, 2, 3)$$

Ans 2. Given to us is a plane $2x+3y+z=6$
and $\vec{F} = x^2 \hat{a}_x + xy \hat{a}_y + z \hat{a}_z$

$$\text{Now } \boxed{z = 6 - 2x - 3y}$$

We know that flux = $\iint_S \vec{D} \cdot \vec{\partial S}$

$$\vec{\partial S} = \frac{2\hat{a}_x + 3\hat{a}_y + \hat{a}_z}{\sqrt{14}}$$

$$\text{Now our } \vec{F} = x^2 \hat{a}_x + xy \hat{a}_y + (6 - 2x - 3y) \hat{a}_z$$

$$\text{Flux} = \frac{1}{\sqrt{14}} \iint_S (x^2 + 3xy + 6 - 2x - 3y) dy dx \quad [At z=0]$$

$$= \frac{1}{\sqrt{14}} \int_{x=0}^3 \int_{y=0}^{2-\frac{2x}{3}} (2x^2 + 3xy + 6 - 2x - 3y) dy dx$$

On solving the above integral we get $\boxed{\text{Flux} = 5.21 \text{ units}}$

Ans 3 @ Given to us is $\int = \frac{k}{r^2} (a \leq r \leq b)$

From Gauss law we know that

$$\int E \cdot dA = \frac{Q_{enc}}{\epsilon_0}$$

Now when $r < a$

$$\Rightarrow Q_{enc} = 0 \Rightarrow \boxed{E = 0}$$

(b) $r \in (a, b)$

$$\begin{aligned} \text{Now } Q_{enc}(r) &= \int_0^{2\pi} \int_0^\pi \int_a^r \frac{k}{r^2} r^2 \sin \theta \, dr \, d\theta \, d\phi \\ &= k(r-a) \times 4\pi \end{aligned}$$

$\Rightarrow$ From Gauss's law

$$E \cdot 4\pi r^2 = \frac{k(r-a) \times 4\pi}{\epsilon_0}$$

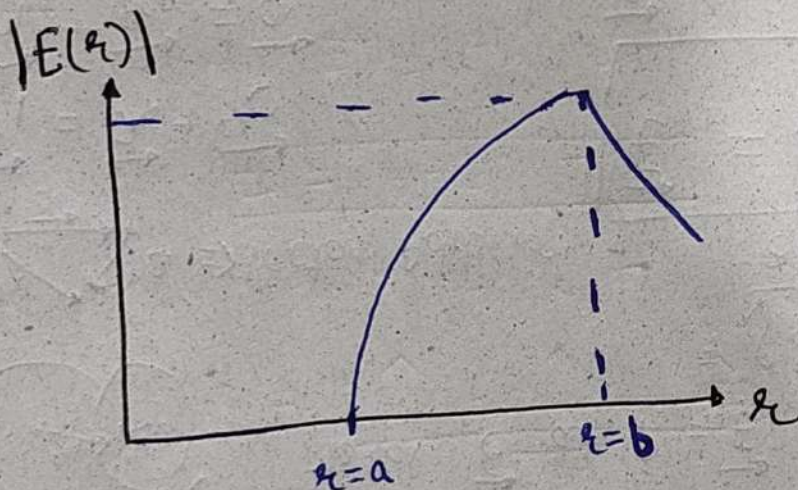
$$\epsilon = \frac{k(r-a)}{\epsilon_0 r^2}$$

③ $r > b$

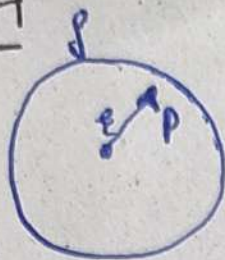
$$Q_{enc} = k(b-a) 4\pi$$

$$\Rightarrow \epsilon \times 4\pi r^2 = 4\pi k(b-a) / \epsilon_0$$

$$\Rightarrow \epsilon = \frac{k(b-a)}{\epsilon_0 r^2}$$



Ans 4

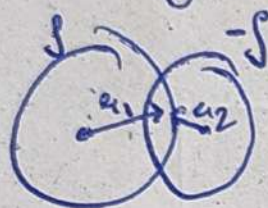


For Point P which is inside of sphere and at distance r

$$\vec{E} \text{ at } P: \epsilon \cdot \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{1}{3} \frac{\rho \cdot \frac{4}{3}\pi r^3}{\epsilon_0}$$

$$\Rightarrow \boxed{\vec{E} = \frac{\rho \vec{r}}{3\epsilon_0}}$$

Now if two spheres with opposite charge density:



$$\text{Total } \vec{E} \text{ at } P: = \vec{E}_1 + \vec{E}_2$$

$$= \frac{\rho \vec{r}_1}{3\epsilon_0} + \frac{-\rho \vec{r}_2}{3\epsilon_0}$$

$$\vec{r}_1, \vec{r}_2 \Rightarrow \text{Vector Addition}$$

$$\vec{r}_1 = \vec{a} + \vec{r}_2$$

$$\vec{r}_1 - \vec{r}_2 = \vec{a}$$

$$\boxed{\vec{E} = \frac{\rho \vec{a}}{3\epsilon_0}}$$

$\vec{E}$ is constant and also uniform throughout the inside region.

Ans 5 ∇ of $\vec{E}$ should be 0 i.e. $\nabla \times \vec{E} = 0$

$$a) \nabla \times \vec{E} = K \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & 2yz & 3xz \end{vmatrix} = K \begin{bmatrix} (-2y)\hat{i} - (3z)\hat{j} \\ (-x)\hat{k} \end{bmatrix} \neq 0$$

$\therefore$ It is not an electric field

$$\begin{aligned}
 \textcircled{b} \quad \vec{\nabla} \times \vec{E} &= k \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^2 & 2xy+z^2 & 2yz \end{vmatrix} \\
 &= k [i(2z-2z) - 0\hat{j} + 0\hat{k}] \\
 &= 0 \\
 &\Rightarrow \text{It is a valid electric field.}
 \end{aligned}$$

Ans 6 $\textcircled{a}$ $W = -\Delta U$

$$\begin{aligned}
 U_{\text{final}} &= U_{\text{ini}} - \frac{kq^2}{a} - \frac{kq^2}{a} + \frac{kq^2}{\sqrt{2}a} \\
 \Rightarrow -\Delta U &= \frac{kq^2}{a} (2 - 1/\sqrt{2}) = \boxed{\frac{1.29 kq^2}{a} = W}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{b} \quad U_{\text{ini}} &= 0 \quad U_{\text{final}} = U_{12} + U_{13} + U_{14} + U_{23} + U_{24} + U_{34} \\
 &= -\frac{kq^2}{a} + \frac{kq^2}{\sqrt{2}a} - \frac{kq^2}{a} - \frac{kq^2}{a} - \frac{kq^2}{a} + \frac{kq^2}{\sqrt{2}a} \\
 &= \frac{kq^2}{a} [-4 + \sqrt{2}] \\
 \Rightarrow W &= -\Delta U = \boxed{\frac{2.54 kq^2}{a} = \text{Ans}}
 \end{aligned}$$